

Topology PhD Exam, August 2012

Work the following problems and show all work. Support all statements to the best of your ability.

1. Prove that the 3-sphere S^3 is not homeomorphic to the 3-space $\mathbb{R}^3$.
2. Let $A \subset \mathbb{R}^2$ be an infinite countable subspace.
 - (a) Can A be connected?
 - (b) Is $\mathbb{R}^2 \setminus A$ connected?
3. Construct a map $f : T^2 \rightarrow S^2$ of degree 3 where $T^2 = S^1 \times S^1$ is a torus.
4. Show that there is no map $S^2 \rightarrow T^2$ of degree 3.
5. Let GL_n be the space of all invertible real $n \times n$ -matrices. Is GL_n compact? Is it connected?

Answer the following with complete definitions or statements or short proofs.

6. Is the 2-sphere S^2 with k points removed homeomorphic to a topological group for
 - (a) $k = 1$?
 - (b) $k = 2$?
 - (c) $k = 3$?
 - (d) $k = 0$?
7. State the Lefschetz Fixed Point Theorem.
8. Compute the Euler characteristic $\chi(\mathbb{CP}^5 \times \mathbb{RP}^5 \times S^5 \times S^2)$.
9. State the Baire Category Theorem
10. Does every function $f : \mathbb{N} \rightarrow \mathbb{R}$ admit a continuous extension $\bar{f} : \beta\mathbb{N} \rightarrow \mathbb{R}$ to the Stone-Čech compactification?
11. State the Five Lemma.
12. What can you say about the k -th cohomology group of a closed oriented manifold for
 - (a) $k = n$?
 - (b) $k = n - 1$?
13. Does there exist a covering space of the figure eight that has a non-trivial abelian fundamental group?
14. Describe all connected subsets of the real line $\mathbb{R}$.
15. State the Contraction Mapping Theorem.